Naresh Puppala

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https://nareshpuppala246.github.io/-portfolio_me/

Skills

Embedded Platforms & Processors:
Infineon PSoC Edge E84, STM32, ESP32, Cypress PSoC, TI Microcontrollers, Arduino, Raspberry Pi, BeagleBone Black, ARM Cortex-M

Programming Languages:
C, C++, Python, SQL

RTOS & Embedded Software:
FreeRTOS, BSP Development, SDK Integration, Hardware Bring-up, Firmware Development

Communication Protocols:
I²C, SPI, UART, CAN, Modbus, DVP

Wireless & Networking:
Wi-Fi, Bluetooth, BLE, Zigbee, LoRa, GSM, MQTT, HTTP, REST APIs

Development Tools & IDEs:
ModusToolbox, STM32CubeIDE, ESP-IDF, Keil µVision, SEGGER Embedded Studio, Arduino IDE, Eclipse IDE

Version Control & Project Management:
GitHub, Bitbucket, JIRA, SourceTree

Hardware Design & Debugging:
PCB Design, KiCad, EasyEDA, Eagle, J-Link, Oscilloscopes, Logic Analyzers, Signal Generators, Power Profilers, Soldering, Testing & Validation

Embedded AI & Software Frameworks:
DeepCraft Studio, OpenCV, TensorFlow, NumPy, Pandas, PySerial, Flask

Sensors & Peripherals:
Temperature, Humidity, Gas, Light, Ultrasonic, Hall-Effect, IMU, Pressure, GPS/GNSS, Camera, Battery Fuel Gauge, Relays, Motors, Buzzers, LED

Summary

Embedded Systems Engineer with 7+ years of experience designing and developing production-grade firmware for industrial, IoT, and smart-device applications. Experienced across the embedded product development lifecycle, including firmware architecture, BSP and SDK integration, hardware bring-up, real-time systems, system debugging, and optimization. Hands-on expertise in Infineon PSoC Edge E84, STM32, ESP32, edge AI, embedded vision, multi-camera processing, wireless connectivity, and low-power systems.Proven track record of delivering and deploying embedded solutions across 5000+ devices in real-world environments.

Experience

BRBS REALITY — Hyderabad
Firmware Engineer — June 2024-Present

Key Contributions:

- **Designed and developed production-grade embedded firmware** using **C, C++, and Python** across **Infineon PSoC Edge E84, STM32, ESP32, Arduino, and other microcontroller platforms**, implementing RTOS-based multitasking, peripheral integration, real-time data processing, and system-level optimization.
- **Developed edge-AI and embedded vision solutions on the Infineon PSoC Edge E84**, including **multi-camera processing, image capture, JPEG processing, image rendering/display integration, and AI model development and evaluation using DeepCraft Studio**.
- **Engineered sensor, peripheral, and connectivity integrations** covering **GPS/GNSS, GSM, Wi-Fi, Bluetooth/BLE, LoRa, cameras, SPI Flash, battery chargers, and fuel-gauge systems**, enabling reliable data acquisition, communication, and device operation.
- **Implemented data-management and power-optimized firmware** using **external SPI NOR Flash for persistent data storage and logging**, along with charger and fuel-gauge integration for battery-powered embedded products.
- **Performed system-level hardware and firmware debugging, validation, and bring-up**, troubleshooting **UART, SPI, I²C, GPIO, sensor interfaces, wireless communication, and PCB-level issues** to improve product reliability and accelerate prototype validation.

Niltech R&D Pvt.Ltd — Hyderabad
Embedded Engineer — March 2019 to June 2024

- Successfully developed and maintained firmware for various controller platforms, including **ATMEL chips**, **ESP**, **Raspberry Pi-**, and **ARM**, demonstrating adaptability across diverse hardware environments to achieve optimal product performance.

- Led the integration of embedded systems with robotics, contributing to the development of innovative products such as **home automation devices**.
- Designed and implemented firmware for a **Smart Card Payphone System**, deployed across **2500+ units**, leveraging communication protocols such as **I2C**, **SPI**, and **UART** to enhance functionality and improve user experience.
- Contributed to critical hardware design discussions by effectively interpreting circuit schematics and resolving complex hardware/firmware issues to support product development.
- Played a key role in the development and maintenance of a specialized device for the **Indian Army** and **DRDO**, showcasing dedication to high-impact projects requiring precision and reliability.

Education

Guru Nanak Institutions B.Tech
Electronics And Communication Engineering June 2016-July 2019

Kshatriya College of Engineering Diploma
Electronics And Communication Engineering June 2013-April 2016

Projects

4G GSM Based Real Time Smart Card Payphone System
Developed and deployed a 4G GSM-based Smart Card Payment System utilizing RFID cards for secure transaction processing. Led the end-to-end embedded system development using C and Python, including firmware development, hardware integration, communication interfaces, system testing, and deployment. The system was successfully implemented across multiple locations and states, with 5000+ devices deployed in real-world environments